

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech (EC) Examination****November 2015****EC 201.01/EC 201 Digital Electronics & Logic Design****Date: 25.11.2015, Wednesday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the below questions. [07]**

- a. Which of the following IC No is used for two input NOR logic gate? [01]
 (a) 74LS00 (c) 74LS02
 (b) 74LS04 (d) 74LS06
- b. Which of the following uses the least power? [01]
 (a) TTL (c) CMOS
 (b) ECL (d) all use same power
- c. Add 47.25 to 55.75 using the 2's complement method. [01]
- d. Simplify the Boolean function using K-map and draw logic diagram for simplified Function. $F(x,y,z,w)=\sum(0,2,4,5,6,8,9,11,13,14,15)$ [02]
- e. Design a 4x1 Multiplexer using 2x1 Multiplexer. [02]

Q – 2 Answer the below questions. [14]

- a. Reduce following function $F(A,B,C,D) = \sum m(1,3,5,8,9,11,15) + d(2,13)$ [04]
- b. Obtain the set of prime implicants for the Boolean expression $F(A,B,C,D)=\sum m(0,1,6,7,8,9,13,14,15)$ using the tabulation method. [05]

OR

- b. Explain Magnitude comparator logic design with block diagram and proper equations. [05]
- c. Design and draw block diagram for BCD adder. [05]

OR

- c. Design a combinational circuit using a ROM. The circuit accepts a 3 bit number and generates an output binary number equal to the square of the input number. [05]

Q – 3 Answer the below questions. [14]

- a. Implement the given function using Multiplexer. [04]

$$F(a,b,c,d)= \sum(1,3,4,11,12,13,14,15)$$

- b. Design a serial adder using a sequential logic procedure and a JK flip flop. [05]

OR

- b. Design a Mod-6 Asynchronous counter using T flip flops. [05]

- c. Draw and explain functioning of a BCD ripple counter. [05]

OR

- c. Draw and explain functioning of 4 bit universal shift register. [05]

SECTION – II

Q - 4 Answer the question below. [07]

- a. The Gray number of $3A7_{16}$ is [01]
 (a) 10011001100 (c) 1001110100
 (b) 01100110110 (d) 1001100110
- b. $A + A'B + A'B'C + A'B'C'D + \dots =$ [01]
 (a) 1 (c) $A + B + C + \dots$
 (b) 0 (d) $A' + B' + C' + \dots$
- c. Design 3 bit odd parity generator. [01]
- d. What is race around condition? Explain it with suitable flipflop. [02]
- e. What is the difference between characteristic table and excitation table? Write the excitation table of D Flip Flop. [02]

Q – 5 Answer the below questions. [14]

- a. Why NAND and NOR gates are known as universal gates? [04]
- b. Design BCD to Excess 3 Code Converter. [05]

OR

- b. Implement the Full Adder with two half adder. [05]
- c. For given Boolean function, $F(A,B,C) = \sum m(1,4,5,6,7)$ with minimum no of literals. [05]
 Also draw the logic diagram for the using NAND Gate only.

OR

- c. Explain D flip flop to SR flip flop conversion. [05]

Q – 6 Answer the below questions. [14]

- a. Write the design procedure for a sequential logic circuits. [04]
- b. Design the 3 bit binary counter with T flip flop. [05]

OR

- b. What do you mean by Master Slave Flip Flop? Explain clocked master slave JK flip flop. [05]
- c. Design a counter using JK Flip Flop which has repeated sequence of the sequence: 0, 1, 2, 4, 5, 6. [05]

OR

- c. Write a brief note on Johnson Counter. [05]
